



Worldwide Importers Sales & Economic Performance Analysis



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Section 63s



Table of Contents

1.0 Introduction.....	4
2.0 Roles and contribution of team members.....	4
3.0 Problem description.....	5
4.0 Overview of solution.....	5
5.0 Description of data.....	6
6.0 Data Preparation.....	7-9
7.0 Visual Model.....	10
8.0 Dashboard design overview (use figures and diagrams)	11-13
9.0 Implementation (include screen shots)	14-18
10.0 Power BI features used (include screen shots)	18-19
11.0 Problems faced.....	20
12.0 Limitations.....	20
13.0 Future work.....	21
14.0 Conclusions.....	21



Table of figures

Figure 1. Applied Steps in Power Query for DimCity.....	9
Figure 2. Applied Steps in Power Query for DimCustomer.....	9
Figure 3. Applied Steps in Power Query for DimEmployee.....	9
Figure 4. Applied Steps in Power Query for DimDate.....	9
Figure 5. Applied Steps in Power Query for FactSales.....	9
Figure 6. Applied Steps in Power Query for DimStockItem.....	9
Figure 7: Star Schema Relationship Model for Worldwide Importers.....	11
Figure 8: Worldwide Importers Dashboard Overview – Page One.....	12
Figure 9: Worldwide Importers Dashboard Overview – Page Two.....	13
Figure 10: Slicer.....	15
Figure 11: KPI cards.....	15
Figure 12: Sales Performance Trend Area Chart.....	16
Figure 13: Item Level Sales & Profit Analysis Clustered Bar Chart.....	16
Figure 14: Regional Performance Table.....	17
Figure 15: Strategic "What-if" Sliders.....	17
Figure 16: Segmental Distribution Analysis via Donut and Pie Charts.....	17
Figure 17: Profit by State Column Chart.....	18
Figure 18: DAX implementation for calculating sales performance.....	19
Figure 19: DAX implementation for Predictive Profit forecasting.....	19
Figure 20: Configuration of "What-if" parameters for user interaction.....	20
Figure 21: Gap analysis measure between actual and predicted performance.....	20

Table of tables

[Table 1: Roles and contribution of team members].....	4
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1.0 Introduction

Modern businesses produce large volumes of sales and customer-related data on a daily basis. Without effective analytical tools, it can be challenging to convert this raw information into meaningful insights that support strategic decision-making. Business Intelligence (BI) systems help organizations process and analyze data efficiently, enabling them to recognize trends, evaluate performance, and support business operations through interactive visualizations and reports.

This project focuses on analyzing the “World Wide Importers” dataset available on Kaggle using Power BI. The dataset simulates the operations of a wholesale company and includes data about customers, products, orders, sales, employees, suppliers, and financial transactions. The main objective of the project is to examine the dataset, uncover important business insights, and design interactive dashboards that assist in data-driven decision-making.

By conducting this analysis, the project seeks to enhance understanding of sales trends, customer purchasing behavior, and overall operational performance. The use of Power BI dashboards provides a clear and interactive way to present information, allowing users to monitor business activities more effectively and identify insights that can contribute to improving business strategies and organizational performance.

2.0 Roles and Contribution of Team Members

Student Name	Student ID	Contribution
Shouq		
Duha		
Farah		
Maria Alhejji		Dashboard Design and User Interaction
Tala		

Table 1: Roles and contribution of team members



3.0 Problem Description

Organizations today generate and store significant amounts of business data every day. However, extracting useful information from this data remains a major challenge for many companies. In the absence of effective analytical and visualization tools, businesses may find it difficult to evaluate sales activities, analyze customer preferences, monitor inventory operations, and measure overall organizational performance.

Conventional reporting approaches often rely on static reports and spreadsheets that limit the ability of decision-makers to quickly detect important trends, patterns, and operational problems. This limitation can negatively affect business planning, customer satisfaction, sales improvement efforts, and operational management.

To overcome these issues, this project applies Business Intelligence (BI) methods using Power BI to analyze the “Worldwide Importers” dataset. The project aims to convert raw and unorganized data into interactive dashboards and visual reports that provide clear insights into sales performance, customer behavior, product analysis, and other key business metrics. These visualizations help support accurate, efficient, and data-driven decision-making processes.

4.0 Overview of Solution

To address the challenges of managing large-scale wholesale data, we developed a dynamic Power BI analytical system that goes beyond traditional static reporting. Our solution is built on a robust data model that integrates a massive historical dataset of over 26,000 sales records with a live Currency API (freecurrencyapi.com) to provide real-time financial context.

The architecture of this solution follows a Star Schema design which efficiently connects core sales transactions to critical business dimensions like customers, stock items, and geography. We didn't just want to show what happened, we aimed to provide a workspace for strategic planning. Therefore, the dashboard is split into two specialized pages:

1. **Descriptive Analytics:** Focusing on immediate performance tracking through KPIs like total sales, profit, and state-level profitability.
2. **Predictive & Prescriptive Analytics:** This is the core innovation of our project. By implementing "What-if" parameters, we allow users to simulate future scenarios, such as adjusting expected growth percentages or currency fluctuation impacts to see how they would affect predicted profits.

By combining these interactive elements with custom DAX calculations such as our automated sales performance labels, this solution transforms raw information into a proactive tool that helps managers make data-driven decisions in an ever-changing economic environment.



5.0 Description of Data

The dataset utilized in this project is the “Worldwide Importers Dataset” collected from Kaggle. This dataset represents the activities of a fictional wholesale company and includes a variety of connected tables covering different business functions, including customers, products, suppliers, employees, orders, and sales records.

The dataset was chosen because it contains practical and realistic business information that is appropriate for Business Intelligence analysis and visualization through Power BI. In addition, the dataset supports relational data modeling, which helps in analyzing connections between different business components and producing valuable analytical insights.

Main Components of the Dataset

- Customer Data: Includes customer names, customer groups, categories, geographic locations, and purchasing groups.
- Sales and Order Information: Contains invoice records, order dates, quantities purchased, pricing details, and transaction values.
- Product Details: Provides information about products such as product names, brands, colors, package types, and stock-related details.
- Supplier Information: Includes supplier names, supplier classifications, and supplier transaction records.
- Employee and Personnel Data: Contains employee details, contact information, and job-related data.
- Location and Geographic Data: Includes information about cities, countries, delivery regions, and other geographic details.

The dataset consists of several related tables that can be linked together within Power BI to build a relational data model. This structure supports the development of interactive dashboards that help analyse sales trends, customer behaviour, product performance, and business operations.

The main purpose of using this dataset is to examine important business performance indicators and convert raw business data into meaningful visual reports and dashboards that assist organizations in making informed decisions.



6.0 Data Preparation

Before starting the analysis and dashboard creation process, several data cleaning and preprocessing steps were performed using Power Query Editor in Power BI. These steps were necessary to improve data quality, ensure consistency, and prepare the dataset for accurate analysis and visualization.

6.1. Remove irrelevant columns:

- We removed the column **‘Sales Territory’** from the **DimCity** table because it contained the same repeated value in all rows and did not provide useful information for the analysis.
- We removed the columns **‘Valid From’**, **‘Valid To’**, and **‘Lineage Key’** from the **DimCustomer** table because they are technical database fields used for data maintenance rather than business analysis.
- We removed the column **‘Photo’** from the **DimEmployee** table because it contained null values and was unnecessary for employee analysis and dashboard visualization.
- We removed the columns **‘Photo’** and **‘Barcode’** from the **DimStockItem** table because the **‘Photo’** column contained null values, while the **‘Barcode’** column contained both null and non-null values but was not useful for our product and sales analysis.

6.2. Handle missing values and blanks:

- In the **‘Credit Limit’** column, we replaced symbols such as **‘?’** and **‘-’** with appropriate numeric values to improve data quality and ensure accurate analysis.
- In the **‘Buying Group’**, **‘Category’**, and **‘Bill To Customer’** columns, we replaced missing and unavailable values such as **‘N/A’** with labels like **‘Other’** and **‘Other Shops’** to maintain data consistency.
- In the **‘Customer’**, **‘Primary Contact’**, and **‘Postal Code’** columns, missing and unknown values were replaced with suitable default values to avoid blank records in the dashboard.
- In the **‘Recommended Retail Price’** column, invalid symbols were removed and missing values were corrected to support accurate product analysis.
- In the **‘Color’** and **‘Brand’** columns, missing values were replaced with default categories such as **‘Blue’** and **‘Other Brands’** to improve completeness of the dataset.



- In the **'Bill To Customer Key'** column, incorrect and missing values were replaced with valid reference values to preserve proper table relationships and reporting accuracy.

6.3. Change data types to match the correct formats:

- We converted columns such as **'City Key'**, **'Customer Key'**, **'Employee Key'**, and **'Stock Item Key'** from Text to Integer to support relationships between tables.
- We changed columns including **'Customer'**, **'City'**, **'Country'**, **'Category'**, and **'Brand'** from General format to Text format for proper categorization and filtering.
- We converted columns such as **'Invoice Date Key'**, **'Delivery Date Key'**, and **'Date'** from Text to Date format to support time-based analysis.
- We changed financial columns including **'Unit Price'**, **'Profit'**, **'Tax Amount'**, and **'Total Including Tax'** from Text to Decimal format for accurate calculations and KPI analysis.
- We converted quantity-related columns such as **'Quantity'**, **'Lead Time Days'**, and **'Total Dry Items'** from Text to Integer format to improve reporting accuracy.
- We checked all column types and corrected them to the proper formats required for Business Intelligence analysis.

6.4. Rename columns to make them more readable and understandable:

- Renamed the column **'Quantity'** to **'Quantity of sold items'** in the **FactSale** table to make the meaning of the column clearer during sales analysis.
- In the **CurrencyRates** table, the column **'Name'** was renamed to **'Currency'** to better represent the currency names stored in the table.
- Renamed the column **'Value'** to **'Rate'** in the **CurrencyRates** table because it represents currency exchange rates and is easier to understand in the dashboard and reports.
- Some column names were simplified and modified to improve readability and make the dataset easier to use during visualization and analysis.

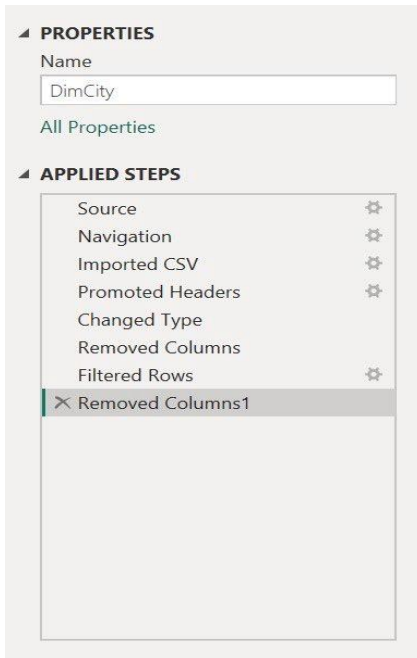


Figure 1

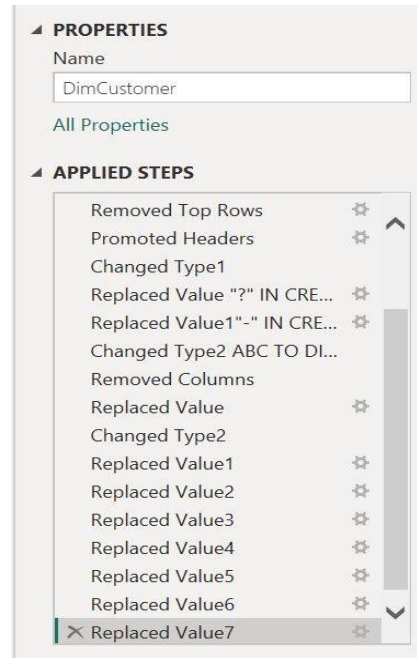


Figure 2

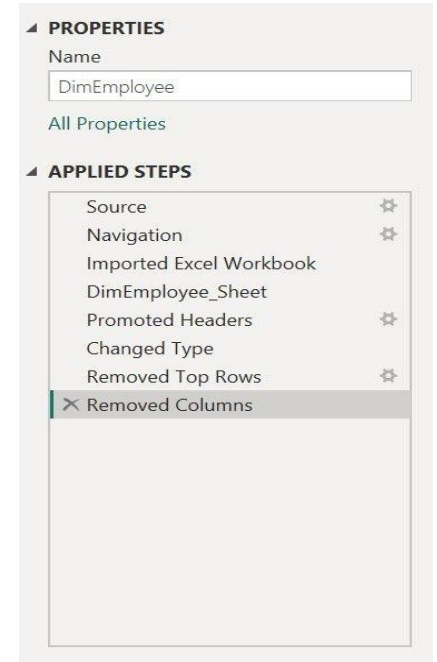


Figure 3

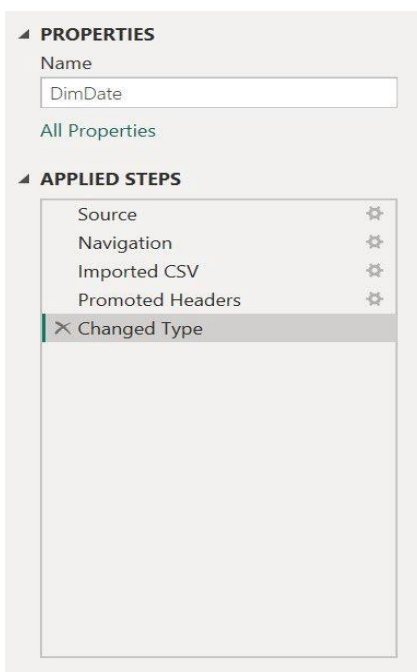


Figure 4

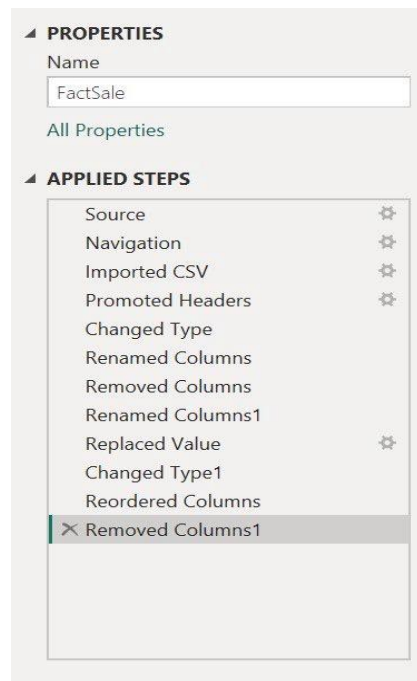


Figure 5

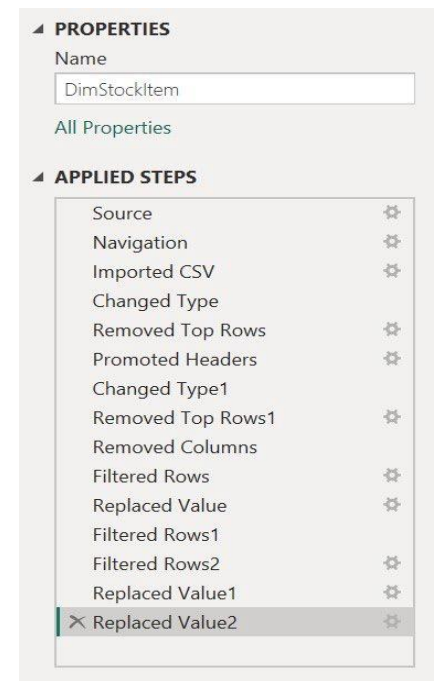


Figure 6



7.0 Visual Model

The data architecture of our project is built on a **Star Schema**, which is a widely recognized best practice for optimizing query performance and ensuring data clarity in Power BI. As illustrated in the relationship view **Figure 7**, *our* model is designed to handle complex filtering across multiple business dimensions without compromising speed or accuracy.

At the heart of the model is the **FactSale table**, which serves as the central hub containing all quantitative transactional data, including more than 26,000 sales records. This fact table is connected via "One-to-Many" (1:*) relationships to several key **Dimension tables**:

- **DimCustomer and DimStockItem:** These tables allow us to slice sales data by customer categories and specific product attributes.
- **DimDate:** A dedicated calendar table that enables sophisticated time-intelligence calculations and trend analysis over different periods.
- **DimCity and DimEmployee:** These dimensions provide the necessary context for geographic performance tracking and individual salesperson accountability.

Beyond the core transactional schema our model incorporates advanced analytical structures to handle dynamic data:

- **API-Driven Currency Rates:** We integrated a **CurrencyRates table** to allow for dynamic financial conversions based on live exchange rates. In Figure 1, the relationship for this table is intentionally kept as a dotted/limited line; this reflects its specialized role in providing rate values via DAX rather than acting as a standard descriptive dimension.
- **Disconnected Parameter Tables:** For our "What-if" analysis, we implemented independent tables for **Expected Growth %** and **Currency Impact %**. These are **disconnected tables** a strategic design choice where the tables are not linked to the data sources through traditional relationships. This approach allows our DAX measures to capture user inputs from slicers and apply them to simulations without affecting the underlying data integrity [1].
This structured hybrid approach ensures that the model remains scalable, interactive, and easy to navigate for stakeholders.

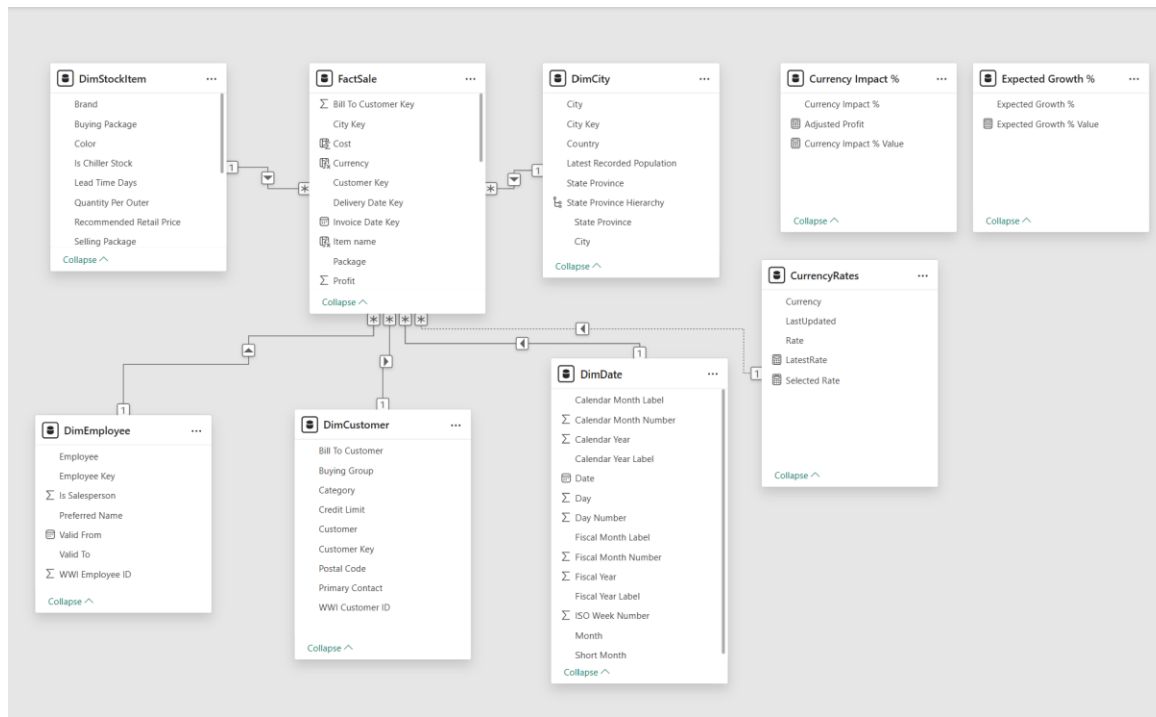


Figure 7: Star Schema Relationship Model for Worldwide Importers

8.0 Dashboard design overview

The Worldwide Importers dashboard is organized into two dynamic pages that allow users to navigate between historical descriptive data and future strategic simulations. The design focuses on providing a clean interface that transforms 26,397 sales records into clear business decisions.

8.1 Key Visual Elements:

- **Executive KPI Scorecard (Page 1):** These KPI cards provide an instant snapshot of the company's health, displaying Total Sales (19.88M), and an overall Profitability rate of 49.92%.
- **Sales Performance Trend (Page 1):** An area chart that visualizes the fluctuation of sales over time, helping managers identify seasonal patterns.
- **State Level Profitability Analysis (Page 1):** A table that uses custom DAX logic to automatically categorize regions into performance tiers (e.g., Exceptional Performance, Average Performance) based on their revenue.
- **Strategic "What-if" Sliders (Page 2):** Interactive sliders for Expected Growth % and Currency Impact that allow stakeholders to simulate different economic scenarios in real-time.
- **Profit Distribution Analysis (Page 2):** A donut chart and column chart that illustrate profit distribution across Buying Groups (like Tailspin Toys) and States, identifying the primary drivers of revenue.



8.2 Design and Aesthetic Choices:

- **Color Theme:** A professional "Sage Green" and "White" palette was selected to create a calm, corporate, and accessible visual experience that enhances data readability.
- **Navigation & Sidebar:** A permanent filter sidebar is positioned on the left, containing all slicers (Year, Employee, Currency, etc.) This ensures consistent navigation and allows users to refine their view without leaving the main analytical area.
- **Interactive Interconnectivity:** All visuals are dynamic and interconnected. Selecting a specific product category or region in one chart will automatically filter all other visuals to show related data segments.
- **Ease of Use:** Designed with a logical hierarchy (from top-level KPIs to detailed tables), the dashboard is easily navigable by users with varying levels of technical expertise, ensuring insights are accessible to everyone.

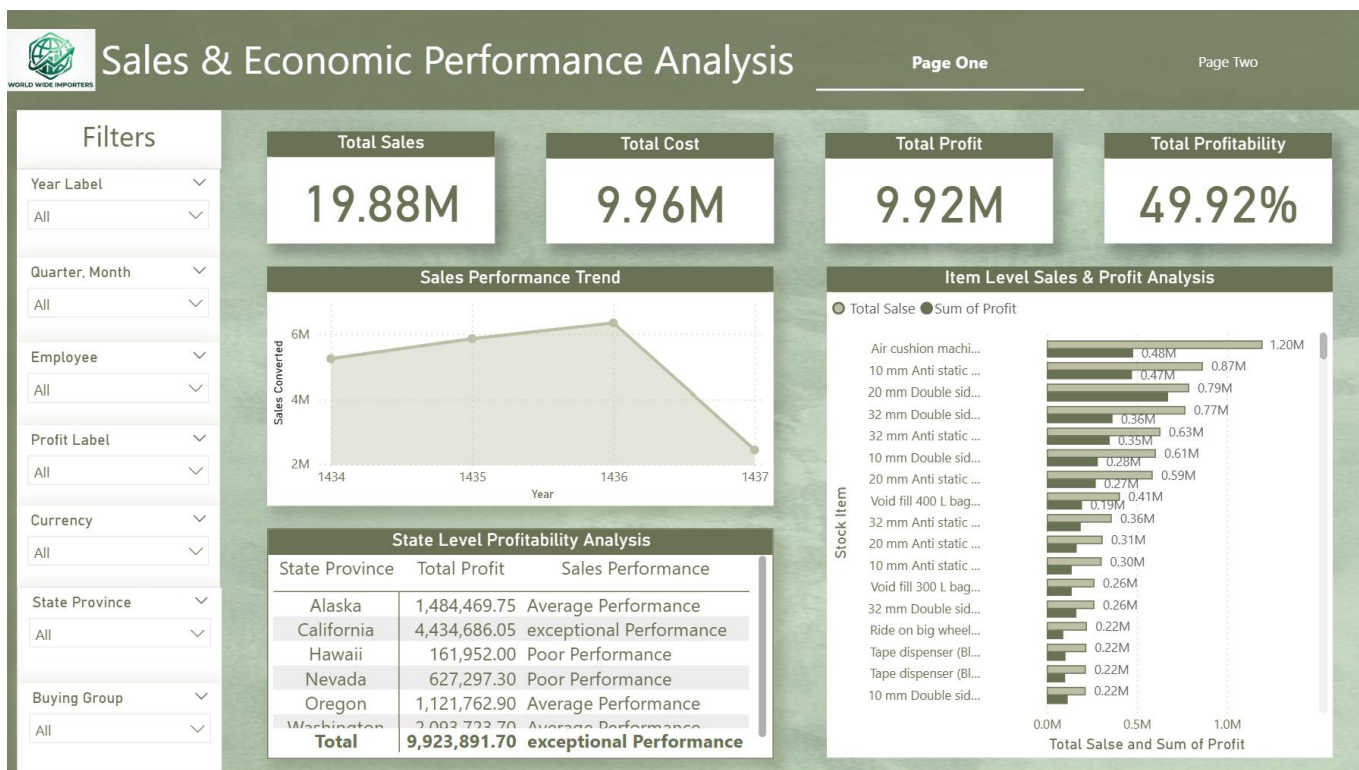


Figure 8: Worldwide Importers Dashboard Overview – Page One

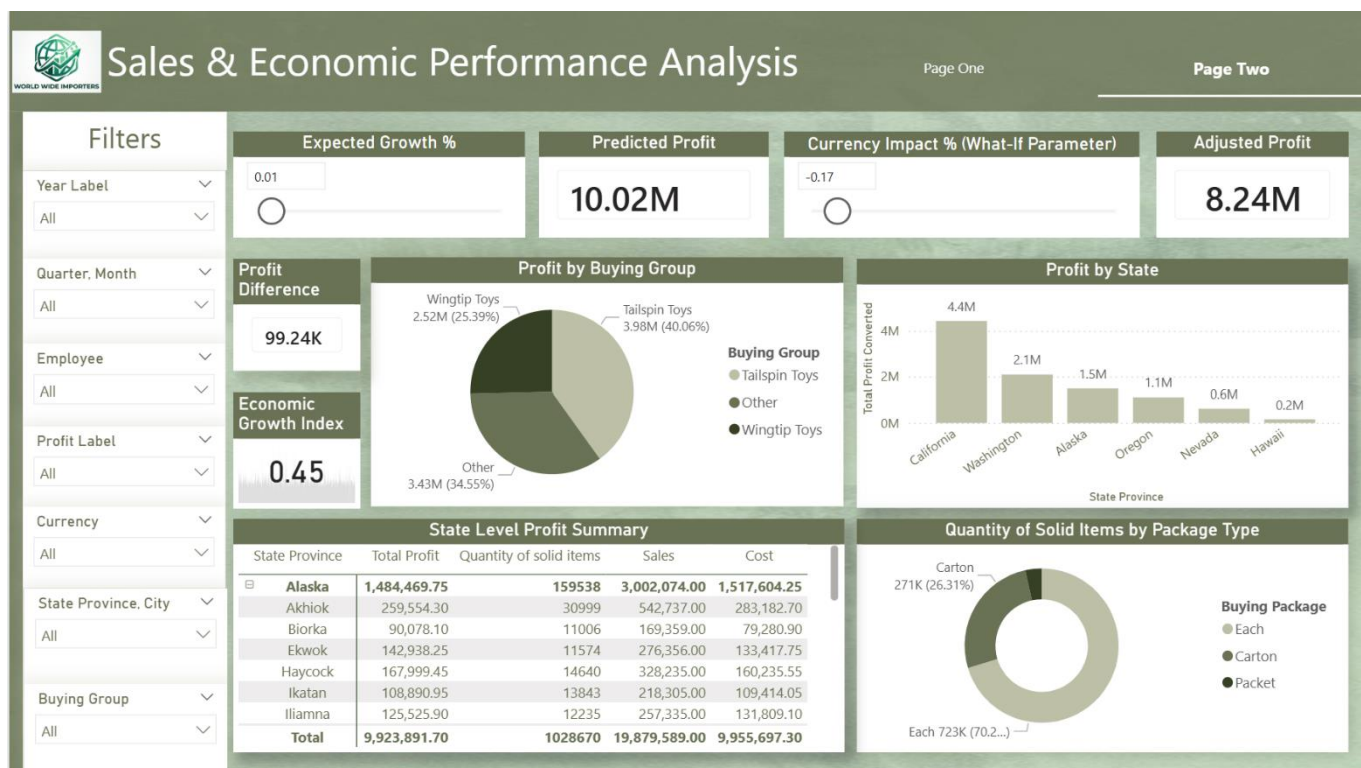


Figure 9: Worldwide Importers Dashboard Overview – Page Two



9.0 Implementation

9.1 Measures:

- **Total Sales, Cost, and Profit:** Calculated using the SUM function on the FactSale table to aggregate all 26,397 sales records into core financial metrics.
- **Total Profitability Converted:** Developed using the DIVIDE function to calculate the percentage of profit relative to sales, ensuring accuracy even when currency rates change.
- **Predicted Profit:** A predictive measure that estimates future earnings based on the growth parameter selected by the user.

*Formula: [Total Profit Converted] * (1 + 'Expected Growth %'[Expected Growth % Value])*

- **Sales Performance Labels:** A prescriptive measure using the SWITCH function to automatically tag regions like "Poor," "Average," "Excellent," or "Exceptional" based on specific revenue thresholds.
- **Profit Difference:** A variance measure used for gap analysis, calculating the difference between the actual profit and the simulated predicted profit.

9.2 Dashboard Layout:

The dashboard is designed with a logical flow to support efficient decision-making. The title "Sales & Economic Performance Analysis" is centered at the top to maintain focus. To ensure high interactivity, a Permanent Filter Sidebar is placed on the left, containing all slicers for easy data drilling. The main visualization area is split into two specialized pages: Page One for historical descriptive insights and Page Two for predictive strategy simulations.



9.3 Visuals:

- **Slicers (Sidebar):** Interactive visual filters allow users to explore data under specific conditions. Users can adjust filters based on Year, Quarter, Employee, Profit Label, Currency, State, and Buying Group.

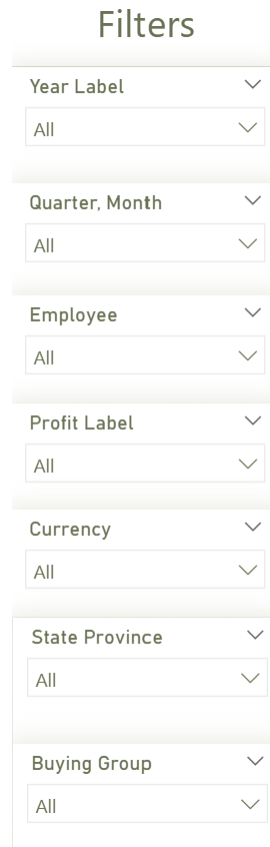


Figure 10: Slicer

- **KPI Cards:** Placed at the top of Page One to provide an immediate snapshot of the company's health, displaying Total Sales (19.88M), Total Profit (\$9.92M), and Total Profitability (49.92%).



Figure 11: KPI cards



- **Area Chart (Sales Performance Trend):** Visualizes the fluctuation of "Sales Converted" over time (Fiscal Years 1434-1437), allowing stakeholders to identify seasonal peaks.

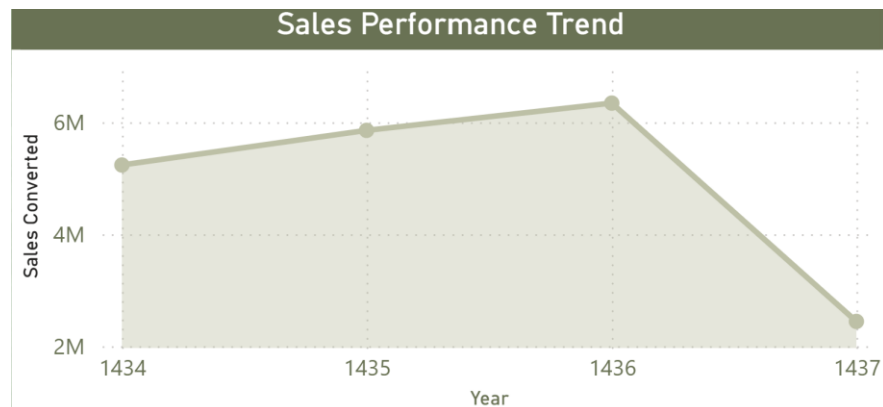


Figure 12: Sales Performance Trend Area Chart

- **Clustered Bar Chart (Item Level Sales & Profit Analysis):** Compares total sales and profit for each stock item. This helps to identify the most profitable products.

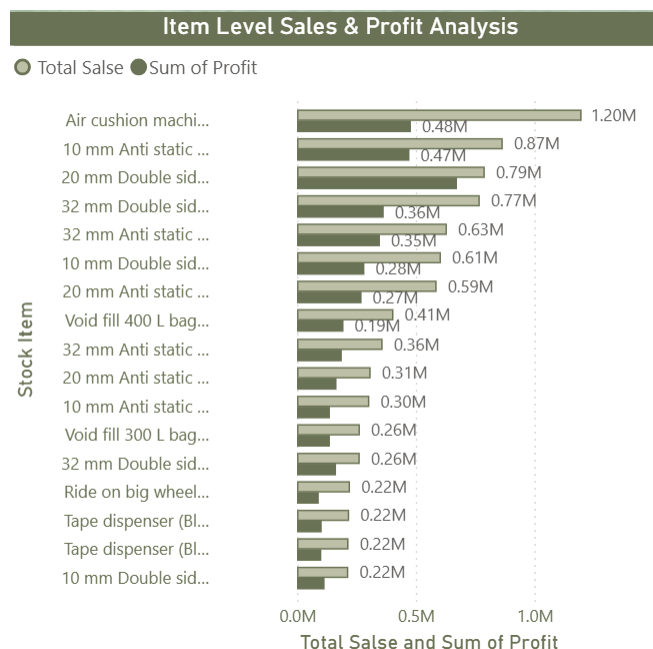


Figure 13: Item Level Sales & Profit Analysis Clustered Bar Chart



- **Performance Table (State Level Profitability Analysis):** A detailed list showing "Total Profit" and "Sales Performance" labels for each state using DAX logic.

State Level Profitability Analysis		
State Province	Total Profit	Sales Performance
Alaska	1,484,469.75	Average Performance
California	4,434,686.05	exceptional Performance
Hawaii	161,952.00	Poor Performance
Nevada	627,297.30	Poor Performance
Oregon	1,121,762.90	Average Performance
Washington	2,093,723.70	Average Performance
Total	9,923,891.70	exceptional Performance

Figure 14: Regional Performance Table

- **Strategic Sliders (What-if Parameters):** Located on Page Two, these interactive sliders for Expected Growth % and Currency Impact act as inputs for our predictive models.

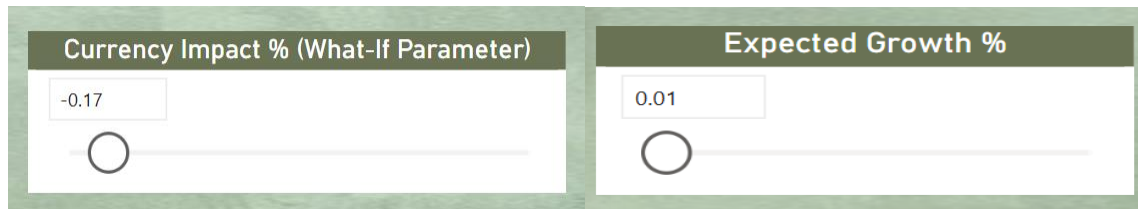


Figure 15: Strategic "What-if" Sliders

- **Donut and Pie Charts (Distribution Analysis):** Analyzes profit distribution by Buying Group and shipping efficiency by Package Type (Carton, Case, and Packet).

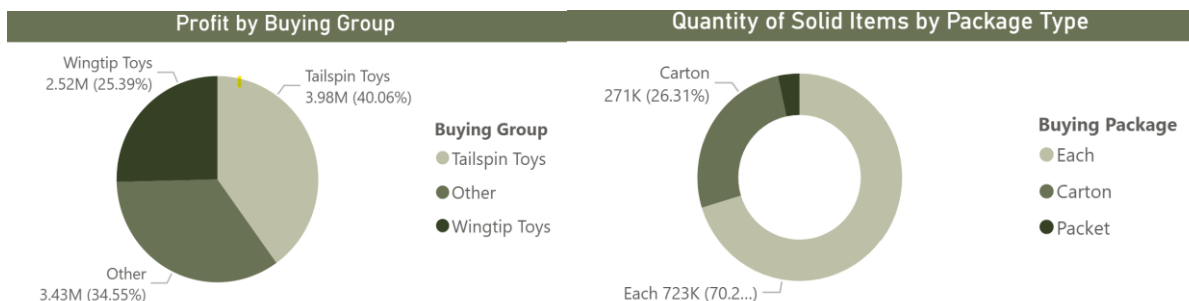


Figure 16: Segmental Distribution Analysis via Donut and Pie Charts



- **Column Chart (Profit by State):** Located on page two, these visual displays total profit distribution across states. It identifies California and Washington as the primary revenue drivers, enabling management to target high-return regions and allocate resources effectively for growth.

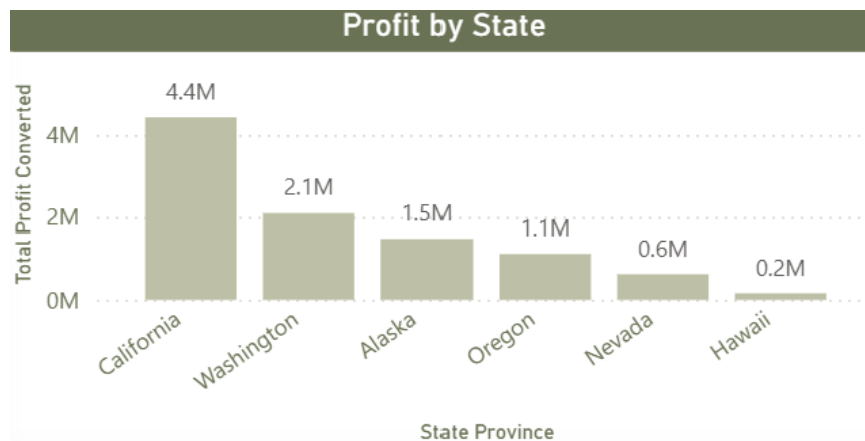


Figure17: Profit by State Column Chart

10.0 Power BI features

Power BI features used to transform the raw data into a strategic decision-support system we utilized several advanced Power BI features that showcase high technical proficiency and analytical depth.

Our approach focused on creating a dynamic environment where users can move beyond static observations to active forecasting.

10.1 Advanced DAX and Custom Metrics We developed a robust library of custom measures using Data Analysis Expressions (DAX) to derive meaningful KPIs.

- **Dynamic Financial Conversion:** Since Worldwide Importers deal with international contexts, we created measures such as Total Profit Converted and Sales Converted. These measures utilize a Selected Rate from our live currency table, ensuring that all financial figures are accurate and up to date.
- **Performance Classification:** As shown in **Figure 15**, we implemented logic-based measures such as Sales Performance, using the SWITCH function to automatically categorize sales results into Poor, Average, Excellent or Exceptional performance tiers. This provides immediate visual cues for managers to identify areas needing attention.



10.2 Predictive Analytics through What-if Analysis The core innovation of our dashboard is the implementation of "What-if" Parameters which transition the report from descriptive to predictive and prescriptive analytics.

- **Profit Forecasting:** We created the Expected Growth parameter which is linked to our Predicted Profit measure:

*Formula: Predicted Profit = [Total Profit Converted] * (1 + 'Expected Growth %'[Expected Growth % Value])*

This feature allows managers to manually adjust growth expectations via a slicer and instantly visualize the projected profit outcomes on the dashboard.

- **Economic Sensitivity Testing:** We also introduced a Currency Impact parameter. This allows users to simulate various economic scenarios such as currency devaluations or appreciations to see their direct effect on the Adjusted Profit.
- **Variance Analysis:** To provide deeper insights we developed the Profit Difference measure. This calculates the gap between current actual and predicted targets, helping stakeholders understand the feasibility of their strategic goals.

Interactive User Engagement To ensure the dashboard is user-friendly and accessible, we incorporated several interactive elements:

- **Integrated Slicers:** Users can filter the entire dataset by Year, Quarter, Month, Employee, Profit Label, Currency, State Province, and Buying Group, providing a highly customized view of the data.
- **Interactive Visualizations:** We utilized a variety of charts including performance tables for state-level profitability and donut charts for buying group analysis, all of which respond instantly to user-defined parameters

```
1 Sales Performance = SWITCH(TRUE,SUM(FactSale[Total Excluding Tax])< 2000000,"Poor Performance",
2 SUM(FactSale[Total Excluding Tax])<5000000, "Average Performance",
3 SUM(FactSale[Total Excluding Tax])<8000000, "Excellent Performance",
4 "exceptional Performance")
```

Figure 18: DAX implementation for calculating sales performance

```
1 Predicted Profit = [Total Profit Converted] * (1 + 'Expected Growth %'[Expected Growth % Value])
```

Figure 19: DAX implementation for Predictive Profit forecasting

```
1 Expected Growth % Value = SELECTEDVALUE('Expected Growth %'[Expected Growth %], 0.05)
```

Figure 20: Configuration of "What-if" parameters for user interaction



$$1 \text{ Profit Difference} = [\text{Predicted Profit}] - [\text{Total Profit Converted}]$$

Figure 21: Gap analysis measure between actual and predicted performance

11. Problems faced

During the development of this project, we faced several challenges while working with the data and building the dashboard.

11.1 Difficulty to finding suitable API

Identifying an appropriate API that aligned with the existing dataset proved to be a time intensive process, Multiple API options were evaluated and test before determining a suitable solution that met the project requirements.

11.2 API integration issue

Integrating the selected API with the established database tables presented considerable challenges, The API data structure didn't align with the existing dataset format, particularly regarding currency representation. This issue was resolved through the implementation of a new column to standardize currency formats and ensure proper data alignment.

11.3 Data preparation challenges

The data preparation phase revealed multiple complications, Including data quality deficiencies, incorrect data types and extraneous columns. These challenges necessitated comprehensive data cleaning procedures and structural adjustments to ensure the data-maintained accuracy, consistency and analytical suitability.

12 Limitations

Even though the project was completed successfully, there are a few limitations that may affect the results and performance of the dashboard.

12.1 Limited Data Recency

One limitation is that the dataset is a bit outdated, as it only includes data up to 2016. Because of this, the analysis may not fully reflect current trends or recent changes.

12.2 API Dependency

The dashboard also relies on external API data, which means there could be delays, limited access, or unexpected changes that are outside our control.



12.3 Performance Limitation

Another limitation is related to performance, as working with large amounts of data and multiple visuals can sometimes slow down the dashboard and affect its responsiveness.

13. Future work

For future work, there are some possible improvements that can be made to enhance the dashboard and provide better insights.

13.1 Using More Recent Data

The dataset can be updated with more recent data to better reflect current trends and improve the accuracy of the analysis.

13.2 Improving Data Refresh Process

Improving the data refresh process can help ensure that the dashboard reflects more up-to-date information and provides more accurate insights.

13.3 Expanding Data Sources

Additional data sources can be integrated to provide a more comprehensive view and enhance the overall analysis.

References

[1] **Power BI Consulting**, “What-If Parameters: Scenario Analysis Guide,” *Power BI Consulting*, 2026. Accessed: May 17, 2026. [Online]. Available: <https://powerbiconsulting.com/blog/power-bi-what-if-parameters-scenario-analysis-2026>

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Appendix A : Read me File for the Report

Appendix B : Read me File for the Dashboard